

# Milestone 1: Data Pipeline and Initial Model

## 1 Dataset Generation Pipeline

### 1.1 Overview

A synthetic dataset was generated for AI/ML tutoring. The dataset is used for:

- Training a difficulty classifier
- Providing examples for retrieval-based tutoring

### 1.2 Dataset Structure

The dataset includes:

- 25 AI/ML topics
- 3 levels: beginner, intermediate, advanced
- 12 examples per topic-level

Total dataset size:

$$25 \times 3 \times 12 = 900$$

Each row contains:

- Question
- Answer
- Level
- Topic
- Angle

*Note: The dataset will be further expanded by increasing samples, topics, and diversity to improve model generalization and retrieval quality.*

## 1.3 Pipeline Implementation

The dataset was generated using a Large Language Model via API.

Pipeline steps:

1. Select topic and level
2. Generate question using prompt engineering
3. Apply quality checks
4. Generate answer
5. Validate answer
6. Save to CSV file

## 1.4 Key Features

### 1.4.1 1. Question Angles

To ensure diversity, multiple angles were used such as:

- Definition
- Real-world example
- Implementation detail
- Research-level question

### 1.4.2 2. Quality Control

The pipeline includes:

- Minimum length check
- Duplicate detection
- Similarity filtering
- Removal of invalid formatting

### 1.4.3 3. Temperature Settings

- Question generation: 0.9 (diversity)
- Answer generation: 0.7 (clarity)

### 1.4.4 4. Resume Capability

If interrupted, the pipeline resumes using the existing dataset file.

## 2 Initial Model Implementation

### 2.1 Model Used

The system uses **DistilBERT** for difficulty classification.

### 2.2 Functionality

- Input: User question
- Output: Predicted level (Beginner / Intermediate / Advanced)

### 2.3 Working

The model learns patterns such as:

- Complexity of language
- Presence of technical terms
- Type of question (definition vs derivation)

### 2.4 Preliminary Results

- Model successfully differentiates basic and advanced questions
- Dataset supports structured classification
- Further tuning is required for higher accuracy

## 3 Conclusion

The data pipeline is fully functional and capable of generating high-quality datasets. The initial model is successfully running and demonstrates the feasibility of adaptive difficulty classification.